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The acoustic boundary layers: a detailed analysis

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Abstract

The viscosity of the medium plays an important role in defining the characteristics of sound wave propagation in ducts. This effect, due to the difficulty of analysis, has been either completely neglected in the literature or considered only approximately.

In this study, a mathematical model is constructed to describe the physical problem in its general form without imposing assumptions a priori. The physical quantities have been conveniently expressed in only two nondimensional quantities, a frequency or wave number bK and a Reynolds number Re . The set of equations describing the model is solved in two-dimensional ducts in the case of no mean flow. Closed-form, exact solutions for the acoustic parameters are obtained. The effect of both Re and bK are computed and presented graphically. The results confirm the existence of a small region, the acoustic boundary layer, inside which the acoustic parameters vary very rapidly. For the first time, the values of u , v and p are computed inside the acoustic boundary layer. The profiles of these parameters are presented across the duct width. The widely used assumption that the viscous effect is confined to the boundary layers, and the rest of the field is essentially inviscid is not supported by the current results.

Key words: Acoustic boundary layers; Duct acoustics; Viscosity effect; Propagation and attenuation of sound waves

1. Introduction

Viscosity plays an important role in the field of duct acoustics. The field of duct acoustics concerns itself with the study of the propagation and attenuation characteristics of sound waves in ducts in all cases of the presence or absence of a mean flow.

The viscosity of the medium inside the duct imposes a no-slip condition at the duct boundary. Moreover, viscosity of the medium together with its thermal conductivity are responsible for the natural attenuation of the sound waves.

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The study of the effect of viscosity on sound attenuation in ducts started in 1868 by the classical work of Kirchhoff [6], who studied the attenuation of the “plane waves” in tubes. Kirchhoff’s work was logically extended by Rayleigh [13] to include higher-order modes.

During the early 1950s, the problem of the effect of viscosity and heat conductivity on the acoustic wave propagation in circular, rectangular and two-dimensional ducts that carry no mean flow received considerable attention.

Extension of Rayleigh’s work, alternative formulation of the problem and experimental results appeared in the literature of that period ([1,2,5,7,15] are good examples).

Later, during the 1960s and early 1970s, the interest focused on the solution of the duct acoustic problem in the presence of mean flow in the duct. The interest in this area was promoted by the need to reduce the aircrafts’ engine noise that became a major public and governmental concern.

Mikhail and Abdelhamid [8,9] studied sound propagation in lined ducts (i.e., ducts with finite admittance at the walls) in the presence of viscous mean flow.

The excellent review [12] refers to over 150 references and covers all aspects of duct acoustics.

In the bulk of the literature so far, the effect of viscosity on the acoustics in ducts that carry mean flow was limited to its effect on the velocity profile of the mean flow. The acoustic disturbance was considered inviscid with the unrealistic slip along the duct boundaries. The only study that considered viscous acoustics disturbance in the presence of mean flow is [11]. The study employs the “acoustic boundary layer concept” that was first introduced by Cremer [3] in 1948. The acoustic boundary layer concept is based on the assumption that most viscosity effects exist in a thin region near the wall. Outside that region, the acoustic disturbance is mainly inviscid. According to this assumption, the viscous acoustic case in a hard duct is treated as an inviscid case in a duct of finite admittance at the wall. This admittance is introduced by and depends on the medium viscosity.

The approach taken in the current research is the formulation of the problem in its general form to include all variables without imposing assumptions a priori. The goal is to provide depth to the understanding of the transmission phenomena and to examine the role played by the different parameters and their effects on the characteristics of the acoustic waves.

In the present study, a two-dimensional duct is considered. The duct consists of two parallel, infinite, rigid planes at a distance $2b$ as shown in Fig. 1.

The choice of this particular geometry provides simplicity in setting the boundary conditions; meanwhile, all the trends of the effect of the different parameters would be representative of any other duct geometry as experience in this field indicates.

The physical quantities have been expressed conveniently in only two nondimensional quantities, a frequency number bK and a Reynolds number Re . The effect of these two

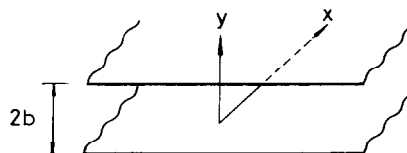


Fig. 1. Duct geometry.

parameters on the characteristics of the acoustic waves have been investigated for zeroth- and higher-order symmetric and anti-symmetric modes.

The effect of viscosity on the propagation and attenuation numbers for a wide range of bK and Re have been reported in [10].

In this paper, the attention is focused on the profile of the acoustic quantities: the acoustic pressure, the longitudinal and transverse particle velocities, in particular, the variation of these quantities inside the acoustic boundary layers and the effect of viscosity on these profiles at a wide range of frequencies.

2. Analytical model

2.1. Basic equations

The flow field inside the duct is governed by five basic equations [14].

(1) *Continuity equation*:

$$\frac{\partial \rho}{\partial t} + \frac{\partial}{\partial x}(\rho u) + \frac{\partial}{\partial y}(\rho v) = 0, \quad (1)$$

where ρ is the density, t is the time, u and v are the velocities in the x - and y -directions respectively.

(2) *Navier–Stokes equation*: x -component of the momentum:

$$\rho \left(\frac{\partial u}{\partial t} + u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} \right) = -\frac{\partial p}{\partial x} + \frac{\partial}{\partial x} \left(2\mu \frac{\partial u}{\partial x} - \frac{2}{3}\mu \nabla \cdot \boldsymbol{\omega} \right) + \frac{\partial}{\partial y} \left(\mu \frac{\partial u}{\partial y} + \mu \frac{\partial v}{\partial x} \right); \quad (2)$$

y -component of the momentum:

$$\rho \left(\frac{\partial v}{\partial t} + u \frac{\partial v}{\partial x} + v \frac{\partial v}{\partial y} \right) = -\frac{\partial p}{\partial y} + \frac{\partial}{\partial y} \left(2\mu \frac{\partial v}{\partial y} - \frac{2}{3}\mu \nabla \cdot \boldsymbol{\omega} \right) + \frac{\partial}{\partial x} \left(\mu \frac{\partial u}{\partial y} + \mu \frac{\partial v}{\partial x} \right), \quad (3)$$

where p is the pressure, μ is the viscosity, $\boldsymbol{\omega}$ is the velocity vector and $\nabla \cdot \boldsymbol{\omega} \equiv \partial u / \partial x + \partial v / \partial y$.

(3) *Energy equation*: the energy equation for a perfect gas can be written as

$$\rho C_p \frac{dT}{dt} = \frac{\partial p}{\partial t} + \left\{ \frac{\partial}{\partial x} \left(k' \frac{\partial T}{\partial x} \right) + \frac{\partial}{\partial y} \left(k' \frac{\partial T}{\partial y} \right) \right\} + \mu \phi_{00},$$

where k' is the thermal conductivity of the gas, C_p is the specific heat at constant pressure, T is the temperature and ϕ_{00} is the dissipation function

$$2 \left\{ \left(\frac{\partial u}{\partial x} \right)^2 + \left(\frac{\partial v}{\partial y} \right)^2 \right\} + \left(\frac{\partial v}{\partial x} + \frac{\partial u}{\partial y} \right)^2 - \frac{2}{3} \left(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} \right)^2.$$

In order to focus on the effects of viscosity on the attenuation of the acoustic waves, we choose to neglect the effect of thermal conductivity of the gas, and in what follows we set

$$k' = 0.$$

Then, the energy equation for a perfect nonheat conducting gas would be

$$\rho C_p \frac{dT}{dt} = \frac{dp}{dt} + \mu \phi_{00}. \quad (4)$$

(4) State equation:

$$P = \rho RT, \quad (5)$$

where R is the gas constant.

The steady state (mean) quantities are ρ_0 , U , V , P_0 , T_0 and ϕ_{00} .

2.2. Perturbation of the flow

Assume that an acoustic perturbation is superimposed over the flow, e.g., from a fan, a microphone or any other acoustic source; hence the new variables will be

$$\rho = \rho_0 + \bar{\rho}, \quad (6)$$

$$u = U + \bar{u}, \quad (7)$$

$$v = V + \bar{v}, \quad (8)$$

$$P = P_0 + \bar{P}, \quad (9)$$

$$T = T_0 + \bar{T}, \quad (10)$$

where the overbar denotes acoustic perturbation.

Equations (1)–(5) governing the flow including the acoustic wave components, after regrouping the terms for convenience, can be written as follows.

Equation (1) becomes

$$\left[\frac{\partial \rho_0}{\partial t} + \frac{\partial}{\partial x}(\rho_0 U) + \frac{\partial}{\partial y}(\rho_0 V) \right] + \left[\frac{\partial}{\partial t} \bar{\rho} + \frac{\partial}{\partial x} \rho_0 \bar{u} + \frac{\partial}{\partial x}(\bar{\rho} U) + \frac{\partial}{\partial y}(\rho_0 \bar{v}) + \frac{\partial}{\partial y}(\bar{\rho} V) \right] + \left[\frac{\partial}{\partial x}(\bar{\rho} \bar{u}) + \frac{\partial}{\partial y}(\bar{\rho} \bar{v}) \right] = 0. \quad (11)$$

Equation (2) becomes

$$\begin{aligned} & \left[\rho_0 \left(\frac{\partial U}{\partial t} + U \frac{\partial U}{\partial x} + V \frac{\partial U}{\partial y} \right) + \frac{\partial P_0}{\partial x} - \frac{\partial}{\partial x} \left(2\mu \frac{\partial U}{\partial x} - \frac{2}{3}\mu \left(\frac{\partial U}{\partial x} + \frac{\partial V}{\partial y} \right) \right) \right. \\ & \quad \left. - \frac{\partial}{\partial y} \left(\mu \frac{\partial U}{\partial y} + \mu \frac{\partial V}{\partial x} \right) \right] \\ & + \left[\rho_0 \left(\frac{\partial \bar{u}}{\partial t} + U \frac{\partial \bar{u}}{\partial x} + V \frac{\partial \bar{u}}{\partial y} + \bar{u} \frac{\partial U}{\partial x} + \bar{v} \frac{\partial U}{\partial y} \right) + \bar{\rho} \left(\frac{\partial U}{\partial t} + U \frac{\partial U}{\partial x} + V \frac{\partial U}{\partial y} \right) \right. \\ & \quad \left. + \frac{\partial \bar{P}}{\partial x} - \frac{\partial}{\partial x} \left(2\mu \frac{\partial \bar{u}}{\partial x} - \frac{2}{3}\mu \left(\frac{\partial \bar{u}}{\partial x} + \frac{\partial \bar{v}}{\partial y} \right) \right) - \frac{\partial}{\partial y} \left(\mu \frac{\partial \bar{u}}{\partial y} + \mu \frac{\partial \bar{v}}{\partial x} \right) \right] \\ & + \left[\rho_0 \left(\bar{u} \frac{\partial \bar{u}}{\partial x} + \bar{v} \frac{\partial \bar{u}}{\partial y} \right) + \bar{\rho} \left(\frac{\partial \bar{u}}{\partial t} + U \frac{\partial \bar{u}}{\partial x} + V \frac{\partial \bar{u}}{\partial y} \right) + \bar{u} \frac{\partial U}{\partial x} + \bar{v} \frac{\partial U}{\partial y} + \bar{\rho} \left(\bar{u} \frac{\partial \bar{u}}{\partial x} + \bar{v} \frac{\partial \bar{u}}{\partial y} \right) \right] = 0. \quad (12) \end{aligned}$$

Equation (3) becomes

$$\begin{aligned} & \left[\rho_0 \left(\frac{\partial V}{\partial t} + U \frac{\partial V}{\partial x} + V \frac{\partial V}{\partial y} \right) + \frac{\partial P_0}{\partial y} + \frac{\partial}{\partial y} \left(2\mu \frac{\partial V}{\partial y} - \frac{2}{3}\mu \left(\frac{\partial U}{\partial x} + \frac{\partial V}{\partial y} \right) \right) - \frac{\partial}{\partial x} \left(\mu \frac{\partial U}{\partial y} + \mu \frac{\partial V}{\partial x} \right) \right] \\ & + \left[\rho_0 \left(\frac{\partial \bar{v}}{\partial t} + U \frac{\partial \bar{v}}{\partial x} + V \frac{\partial \bar{v}}{\partial x} + \bar{u} \frac{\partial V}{\partial x} + \bar{v} \frac{\partial V}{\partial y} \right) + \bar{\rho} \left(\frac{\partial V}{\partial t} + U \frac{\partial V}{\partial x} + V \frac{\partial V}{\partial y} \right) + \frac{\partial \bar{P}}{\partial y} \right. \\ & \quad \left. - \frac{\partial}{\partial y} \left(2\mu \frac{\partial \bar{v}}{\partial y} - \frac{2}{3}\mu \left(\frac{\partial \bar{u}}{\partial x} + \frac{\partial \bar{v}}{\partial y} \right) \right) - \frac{\partial}{\partial x} \left(\mu \frac{\partial \bar{u}}{\partial y} + \mu \frac{\partial \bar{v}}{\partial x} \right) \right] \\ & + \left[\rho_0 \left(\bar{u} \frac{\partial \bar{v}}{\partial x} + \bar{v} \frac{\partial \bar{v}}{\partial y} \right) + \bar{\rho} \left(\frac{\partial \bar{v}}{\partial t} + \bar{u} \frac{\partial V}{\partial x} + \bar{v} \frac{\partial V}{\partial y} + \bar{u} \frac{\partial \bar{v}}{\partial x} + \bar{v} \frac{\partial \bar{v}}{\partial y} + V \frac{\partial \bar{v}}{\partial y} + U \frac{\partial \bar{v}}{\partial x} \right) \right] = 0. \end{aligned} \quad (13)$$

Equation (4) becomes

$$\left[\rho_0 C_p \frac{dT_0}{dt} - \frac{dP_0}{dt} - \mu \Phi_{00} \right] + \left[\rho_0 C_p \frac{d\bar{T}}{dt} + \bar{\rho} C_p \frac{dT_0}{dt} - \frac{d\bar{P}}{dt} - \mu \bar{\Phi}_{00} \right] = 0. \quad (14)$$

Equation (5) becomes

$$[P_0 - \rho_0 R T_0] + [\bar{P} - \rho_0 R \bar{T} - \bar{\rho} R T_0] - [(\bar{\rho} R \bar{T})] = 0. \quad (15)$$

Simplify equations (11)–(15) taking into account that

(i) the equilibrium quantities satisfy the basic equations, hence the expression in the first square brackets in each equation vanishes;

(ii) we only consider a first-order perturbation, hence the product of perturbed quantities will be neglected, i.e., the expression in third square brackets in each equation will, also, be neglected;

$$(iii) \quad \frac{\partial}{\partial t}(\rho_0, U, T_0) = 0, \quad \frac{\partial}{\partial y} \rho_0 = \frac{\partial}{\partial x} \rho_0 = 0;$$

(iv) the mean flow is parallel, i.e., $V = 0$;

(v) $U = U(Y)$;

(vi) $\mu = \text{constant}$.

Further, let us consider harmonic variation in time so that the basic acoustic quantities can be written as

$$\bar{\rho} = \rho(y) \exp(i\omega t - \gamma x), \quad (16)$$

$$\bar{u} = u(y) \exp(i\omega t - \gamma x), \quad (17)$$

$$\bar{v} = v(y) \exp(i\omega t - \gamma x), \quad (18)$$

$$\bar{P} = P(y) \exp(i\omega t - \gamma x), \quad (19)$$

$$\bar{T} = T(y) \exp(i\omega t - \gamma x), \quad (20)$$

where ω is the acoustic frequency, γ is the complex wave number $\alpha' + i\beta'$, α is the attenuation

coefficient, β' is the phase difference coefficient, i is the imaginary number $\sqrt{-1}$. Substituting (16)–(20) in (11)–(15) and dropping the argument y , we get

$$\Omega\rho = \rho_0 \left(\gamma u - \frac{dv}{dy} \right), \quad (21)$$

$$\mu \frac{d^2 u}{dy^2} + \frac{4}{3} \mu \gamma^2 u - \rho_0 \Omega u = \left(\rho_0 \frac{du}{dy} \right) v + \frac{1}{3} \mu \gamma \frac{dv}{dy} - \gamma P, \quad (22)$$

$$\frac{4}{3} \mu \frac{d^2 v}{dy^2} + \mu \gamma^2 v - \rho_0 \Omega v = \frac{1}{3} \gamma \mu \frac{du}{dy} + \frac{dP}{dy}, \quad (23)$$

$$\Omega T = \frac{\Omega y}{\rho_0 C_p} P + \frac{2\mu}{\rho_0 C_p} \left(\frac{du}{dy} - \gamma v \right) \frac{dU}{dy} \quad (24)$$

and

$$P = R(\rho_0 T + T_0 \rho), \quad (25)$$

where $\Omega = i\omega - \gamma U$.

2.3. Normalization

It is much more practical to obtain solutions in nondimensional form. For this reason, our variables are expressed in terms of convenient basic geometric and ambient constant quantities.

2.3.1. Basic normalization quantities

Speed: ambient speed of sound $C_a = (\bar{\gamma} R T_a)^{1/2}$, where $\bar{\gamma}$ is the ratio of specific heats C_p/C_v , C_v is the specific heat at constant volume and T_a is the static ambient temperature.

Length: b is the half width of the duct.

Time: b/C_a .

Density: ρ_a is the ambient density.

Pressure: $\rho_a C_a^2$.

Temperature: T_a is the static ambient temperature.

2.3.2. Normalized parameters

Normalized velocities: x -component $U_n = U(y)/C_a$; y -component $V_n = V(y)/C_a$; mean flow $M = U(y)/C_a$ is the Mach number.

Normalized length: $y_n = y/b$.

Normalized density: $\rho_n = \rho(y)/\rho_a$, $\rho_{0n} = \rho_0/\rho_a$.

Reynolds number: $Re = \rho_a C_a b/\mu$.

Normalized pressure: $P_n = P(y)/\rho_a C_a^2$.

Normalized temperature: $T_n = T(y)/T_a$, $T_{0n} = T_0/T_a$ and $\Omega(y) = i\omega - \gamma U = i\omega(1 - kM)$ where $k = \gamma/iK$ and $K = \omega/C_a$. Let $\Omega_n = \Omega(y)b/C_a$; then $\Omega_n = ibK(1 - kM)$.

Note that the presence of the imaginary number i will cause the basic parameters u , v , P , ..., etc. to be complex.

2.3.3. Normalized governing equations

Substituting the normalized parameters in the basic governing equations (21)–(25), we get the five governing equations — after dropping the subscript n — in the form

$$\Omega\rho = \rho_0 \left(\delta u - \frac{dv}{dy} \right), \quad (26)$$

$$\frac{1}{\text{Re}} \frac{d^2 u}{dy^2} + \left(\frac{4}{3} \frac{\delta^2}{\text{Re}} - \rho_0 \Omega \right) u = \rho_0 v \frac{dM}{dy} + \frac{\delta}{3 \text{Re}} \frac{dv}{dy} - \delta P, \quad (27)$$

$$\frac{4}{3 \text{Re}} \frac{d^2 v}{dy^2} + \left(\frac{\delta^2}{\text{Re}} - \rho_0 \Omega \right) v = \frac{\delta}{3 \text{Re}} \frac{du}{dy} + \frac{dP}{dy}, \quad (28)$$

$$\rho_0 \Omega T \frac{1}{(\bar{\gamma} - 1)} = \Omega P + \frac{2}{\text{Re}} \left(\frac{du}{dy} - \delta v \right) \frac{dM}{dy}, \quad (29)$$

$$P = \frac{1}{\bar{\gamma}} (\rho_0 T + T_0 \rho), \quad (30)$$

where

$$\delta = \gamma b = i(bK)k = \delta_1 + i\delta_2. \quad (31)$$

These are the set of five simultaneous differential equations in the five normalized acoustic parameters P , u , v , T and ρ .

The problem can be further reduced by substituting (26) and (29) into (30) to eliminate ρ and T , and we end up with only three equations in the three unknowns P , u and v , as follows:

$$\frac{1}{\text{Re}} \frac{d^2 u}{dy^2} + \left(\frac{4}{3} \frac{\delta^2}{\text{Re}} - \rho_0 \Omega \right) u = \rho_0 v \frac{dM}{dy} + \frac{\delta}{3 \text{Re}} \frac{dv}{dy} - \delta P, \quad (32)$$

$$\frac{4}{3 \text{Re}} \frac{d^2 v}{dy^2} + \left(\frac{\delta^2}{\text{Re}} - \rho_0 \Omega \right) v = \frac{\delta}{3 \text{Re}} \frac{du}{dy} + \frac{dP}{dy}, \quad (33)$$

$$\Omega p = \left\{ \frac{2(\bar{\gamma} - 1)}{\text{Re}} \frac{dM}{dy} \right\} \frac{du}{dy} + \delta u \gamma P_0 - T_0 \frac{dv}{dy} - \left\{ 2 \frac{(\bar{\gamma} - 1)}{\text{Re}} \frac{dM}{dy} \right\} \delta v. \quad (34)$$

3. Solution of the case of zero mean flow

In this case we have

$$M = M_0 = 0, \quad \rho_0 = 1, \quad T_0 = 1;$$

hence $\Omega = ibK(1 - kM) = ibK$. Therefore, (32)–(34) become

$$\frac{1}{\text{Re}} \frac{d^2 u}{dy^2} + \left(\frac{4}{3} \frac{\delta^2}{\text{Re}} - ibK \right) u - \frac{\delta}{3 \text{Re}} \frac{dv}{dy} + \delta P = 0, \quad (35)$$

$$\frac{4}{3 \text{Re}} \frac{d^2 v}{dy^2} + \left(\frac{\delta^2}{\text{Re}} - ibK \right) v - \frac{\delta}{3 \text{Re}} \frac{du}{dy} - \frac{dP}{dy} = 0, \quad (36)$$

$$(ibK)P - \delta u + \frac{dv}{dy} = 0. \quad (37)$$

Equation (37) gives P in terms of u, v as follows:

$$P = \frac{\delta u}{ibK} - \frac{1}{ibK} \frac{dv}{dy}. \quad (38)$$

This, when substituted back in equations (35) and (36), results in two simultaneous equations in u, v as follows:

$$\frac{1}{\text{Re}} \frac{d^2 u}{dy^2} + \left(\frac{4}{3} \frac{\delta^2}{\text{Re}} - ibK + \frac{\delta^2}{ibK} \right) u - \left(\frac{\delta}{3 \text{Re}} + \frac{\delta}{ibK} \right) \frac{dv}{dy} = 0, \quad (39)$$

$$\left(\frac{4}{3 \text{Re}} + \frac{1}{ibK} \right) \frac{d^2 v}{dy^2} - \left(\frac{\delta}{3 \text{Re}} + \frac{\delta}{ibK} \right) \frac{du}{dy} + \left(\frac{\delta^2}{\text{Re}} - ibK \right) v = 0. \quad (40)$$

After a lengthy mathematical manipulation, a solution for u, v is found to be of the form

$$u(y) = A_1 e^{\alpha y} + A_2 e^{-\alpha y} + A_3 e^{\beta y} + A_4 e^{-\beta y} \quad (41)$$

and

$$v(y) = \lambda \{ \psi A_1 e^{\alpha y} - \psi A_2 e^{-\alpha y} + \phi A_3 e^{\beta y} - \phi A_4 e^{-\beta y} \}, \quad (42)$$

where

$$\begin{aligned} \alpha^2 &= \left(\frac{1}{4} C_2^2 - C_0 \right)^{1/2} - \frac{1}{2} C_2, & \beta^2 &= - \left(\frac{1}{4} C_2^2 - C_0 \right)^{1/2} - \frac{1}{2} C_2, \\ C_2 &= \frac{a_2}{a_4}, & C_0 &= \frac{a_0}{a_4}, \\ a_4 &= \frac{4}{3 \text{Re}} + \frac{1}{i \text{Re}(bK)}, & a_2 &= \frac{8\delta^2}{3 \text{Re}^2} + \frac{2\delta^2}{i \text{Re}(bK)} - i \frac{7(bK)}{3 \text{Re}} - 1, \\ a_0 &= \frac{4\delta^2}{3 \text{Re}^2} - i \frac{7\delta^2(bK)}{3 \text{Re}} + \frac{\delta^4}{i \text{Re}(bK)} - (bK)^2 - \delta^2, \\ \lambda &= \left(\frac{\delta}{3 \text{Re}} + \frac{\delta}{ibK} \right)^{-1}, \\ \psi &= \frac{\alpha}{\text{Re}} + \frac{1}{\alpha} \left(\frac{4\delta^2}{3 \text{Re}} - ibK + \frac{\delta^2}{ibK} \right), & \phi &= \frac{\beta}{\text{Re}} + \frac{1}{\beta} \left(\frac{\delta^2}{3 \text{Re}} - ibK + \frac{\delta^2}{ibK} \right). \end{aligned} \quad (43)$$

To get the values of the constants A_1 – A_4 and the value of $\delta = \delta_1 + i\delta_2$, we make use of the boundary conditions

$$u(\pm 1) = 0, \quad v(\pm 1) = 0. \quad (44)$$

Applying (41) and (42) at the boundaries, we get

$$A_1 e^{\alpha} + A_2 e^{-\alpha} + A_3 e^{\beta} + A_4 e^{-\beta} = 0, \quad (45)$$

$$A_1 e^{-\alpha} + A_2 e^{\alpha} + A_3 e^{-\beta} + A_4 e^{\beta} = 0, \quad (46)$$

$$A_1 e^{\alpha} \psi - A_2 e^{-\alpha} \psi + A_3 e^{\beta} \phi - A_4 e^{-\beta} \phi = 0, \quad (47)$$

$$A_1 e^{-\alpha} \psi - A_2 e^{\alpha} \psi + A_3 e^{-\beta} \phi - A_4 e^{\beta} \phi = 0. \quad (48)$$

Hence, we have four homogeneous equations in four unknowns, namely A_1 – A_4 . Therefore, to have a nontrivial solution, the determinant Δ of the coefficients of these unknowns must vanish, where

$$\Delta = \begin{vmatrix} e^\alpha & e^{-\alpha} & e^\beta & e^{-\beta} \\ e^{-\alpha} & e^\alpha & e^{-\beta} & e^\beta \\ e^\alpha \psi & -e^{-\alpha} \psi & e^\beta \phi & -e^{-\beta} \phi \\ e^{-\alpha} \psi & -e^\alpha \psi & e^{-\beta} \phi & -e^\beta \phi \end{vmatrix} = 0. \quad (49)$$

Expanding and simplifying, the above condition can be written as

$$\Delta = \{(\psi - \phi) \sinh(\alpha + \beta) + (\psi + \phi) \sinh(\alpha - \beta)\} \\ \times \{(\psi - \phi) \sinh(\alpha + \beta) - (\psi + \phi) \sinh(\alpha - \beta)\}. \quad (50)$$

Expanding once more, we get

$$\Delta = 4\{\phi \sinh \alpha \cosh \beta - \psi \cosh \alpha \sinh \beta\}\{\psi \sinh \alpha \cosh \beta - \phi \cosh \alpha \sinh \beta\} = 0. \quad (51)$$

Eq. (51) is used to get the value of $\delta = \delta_1 + i\delta_2$, using the Newton–Raphson iteration technique.

Now, to get the values of the unknowns A_1 – A_4 we will consider one of them, e.g. A_1 , as an arbitrary constant; then we obtain the other unknowns, i.e., A_2 – A_4 , in terms of A_1 .

Eliminating A_3 and A_4 from (45)–(48), we get

$$A_2 = -\frac{\psi + \phi \sinh(\alpha - \beta)}{\psi - \phi \sinh(\alpha + \beta)} A_1. \quad (52)$$

But from (50) we have

$$(\psi + \phi) \sinh(\alpha - \beta) = \pm (\psi - \phi) \sinh(\alpha + \beta). \quad (53)$$

Hence, after substituting (53) in (52) we get

$$A_2 = \mp A_1. \quad (54)$$

The sign depends on odd or even symmetry, as will be discussed, in details, later.

Eliminating A_1 and A_2 from (45)–(48), we get

$$A_3 = -\frac{\psi + \phi \sinh(\alpha - \beta)}{\psi - \phi \sinh(\alpha + \beta)} A_4. \quad (55)$$

Hence, using (53) we get

$$A_3 = \mp A_4. \quad (56)$$

Also, the sign depends on the condition of symmetry exactly as in (54). Now, we can conclude that if

$$A_1 = A_2, \quad (57)$$

then

$$A_3 = A_4; \quad (58)$$

and if

$$A_1 = -A_2, \quad (59)$$

then

$$A_3 = -A_4. \quad (60)$$

Hence, after substituting (57) and (58) in (45) we get

$$A_1 e^\alpha + A_1 e^{-\alpha} + A_3 e^\beta + A_3 e^{-\beta} = 0,$$

that is,

$$A_3 = -\frac{\cosh \alpha}{\cosh \beta} A_1 \quad (61)$$

and

$$A_4 = -\frac{\cosh \alpha}{\cosh \beta} A_1. \quad (62)$$

Also, after substituting (59) and (60) in (45) we get

$$A_1 e^\alpha - A_1 e^{-\alpha} + A_3 e^\beta - A_3 e^{-\beta} = 0,$$

that is,

$$A_3 = -\frac{\sinh \alpha}{\sinh \beta} A_1 \quad (63)$$

and

$$A_4 = \frac{\sinh \alpha}{\sinh \beta} A_1. \quad (64)$$

The above results can be summarized as follows.

(I) If $(\psi - \phi) \sinh(\alpha + \beta) + (\psi + \phi) \sinh(\alpha - \beta) = 0$, then

$$A_1 = A_2 \quad \text{and} \quad A_3 = A_4 = -\frac{\cosh \alpha}{\cosh \beta} A_1.$$

(II) If $(\psi - \phi) \sinh(\alpha + \beta) - (\psi + \phi) \sinh(\alpha - \beta) = 0$, then

$$A_1 = -A_2 \quad \text{and} \quad A_3 = -A_4 = -\frac{\sinh \alpha}{\sinh \beta} A_1.$$

Now get the values of the flow parameters $u(y)$, $v(y)$ and $P(y)$ as follows.

(I) $A_1 = A_1$ and $A_3 = A_4$. Since $u(y) = A_1 e^{\alpha y} + A_2 e^{-\alpha y} + A_3 e^{\beta y} + A_4 e^{-\beta y}$, therefore,

$$u(y) = A_1 \left\{ (e^{\alpha y} + e^{-\alpha y}) + \frac{A_3}{A_1} (e^{\beta y} + e^{-\beta y}) \right\} = 2A_1 \left\{ \cosh \alpha y - \frac{\cosh \alpha}{\cosh \beta} \cosh \beta y \right\},$$

that is,

$$u(y) = \frac{2A_1}{\cosh \beta} \{ \cosh \beta \cosh \alpha y - \cosh \alpha \cosh \beta y \}. \quad (65)$$

Also,

$$\begin{aligned} v(y) &= \lambda \{ \psi A_1 e^{\alpha y} - \psi A_2 e^{-\alpha y} + \phi A_3 e^{\beta y} - \phi A_4 e^{-\beta y} \} \\ &= 2\lambda A_1 \left\{ \psi \sinh \alpha y - \phi \frac{\cosh \alpha}{\cosh \beta} \sinh \beta y \right\}, \end{aligned}$$

that is,

$$v(y) = \frac{2\lambda A_1}{\cosh \beta} \{ \psi \cosh \beta \sinh \alpha y - \phi \cosh \alpha \sinh \beta y \}, \quad (66)$$

and

$$Dv(y) = \frac{2\lambda A_1}{\cosh \beta} \{ \psi \alpha \cosh \beta \cosh \alpha y - \phi \beta \cosh \alpha \cosh \beta y \},$$

where $D \equiv d/dy$, the differential operator. Hence,

$$\begin{aligned} p(y) &= \frac{\delta u(y) - Dv(y)}{ibk} \\ &= \frac{1}{ibk} \left\{ \delta \left(\frac{2A_1}{\cosh \beta} \right) [\cosh \beta \cosh \alpha y - \cosh \alpha \cosh \beta y] \right. \\ &\quad \left. - \lambda \frac{2A_1}{\cosh \beta} [\psi \alpha \cosh \beta \cosh \alpha y - \phi \beta \cosh \alpha \cosh \beta y] \right\}, \end{aligned}$$

that is,

$$p(y) = \frac{2A_1}{ibk \cosh \beta} \{ (\delta - \lambda \psi \alpha) \cosh \beta \cosh \alpha y - (\delta - \lambda \phi \beta) \cosh \alpha \cosh \beta y \}. \quad (67)$$

(II) $A_1 = -A_2$ and $A_3 = -A_4$. Since $u(y) = A_1 e^{\alpha y} + A_2 e^{-\alpha y} + A_3 e^{\beta y} + A_4 e^{-\beta y}$, therefore,

$$u(y) = A_1 \left\{ (e^{\alpha y} - e^{-\alpha y}) + \frac{A_3}{A_1} (e^{\beta y} - e^{-\beta y}) \right\} = 2A_1 \left\{ \sinh \alpha y - \frac{\sinh \alpha}{\sinh \beta} \sinh \beta y \right\},$$

that is,

$$u(y) = \frac{2A_1}{\sinh \beta} \{ \sinh \beta \sinh \alpha y - \sinh \alpha \sinh \beta y \}. \quad (68)$$

Also,

$$\begin{aligned} v(y) &= \lambda \{ \psi A_1 e^{\alpha y} - \psi A_2 e^{-\alpha y} + \phi A_3 e^{\beta y} - \phi A_4 e^{-\beta y} \} \\ &= 2\lambda A_1 \left\{ \psi \cosh \alpha y - \phi \frac{\sinh \alpha}{\sinh \beta} \cosh \beta y \right\}, \end{aligned}$$

that is,

$$v(y) = \frac{2\lambda A_1}{\sinh \beta} \{\psi \sinh \beta \cosh \alpha y - \phi \sinh \alpha \cosh \beta y\} \quad (69)$$

and

$$Dv(y) = \frac{2\lambda A_1}{\sinh \beta} \{\psi \alpha \sinh \beta \sinh \alpha y - \phi \beta \sinh \alpha \sinh \beta y\}.$$

Hence,

$$p(y) = \frac{\delta u(y) - Dv(y)}{ibK}.$$

So,

$$p(y) = \frac{2A_1}{ibK \sinh \beta} \{(\delta - \lambda\psi\alpha) \sinh \beta \sinh \alpha y - (\delta - \lambda\phi\beta) \sinh \alpha \sinh \beta y\}. \quad (70)$$

3.1. Conditions of symmetry of the axial and normal velocities

We may think of conditions of symmetry due to symmetry of the geometry of our problem. The longitudinal velocity $u(y)$ is given by

$$u(y) = A_1 e^{\alpha y} + A_2 e^{-\alpha y} + A_3 e^{\beta y} + A_4 e^{-\beta y}, \quad (41)$$

and the normal velocity $v(y)$ is given by

$$v(y) = \lambda \{\psi A_1 e^{\alpha y} - \psi A_2 e^{-\alpha y} + \phi A_3 e^{\beta y} - \phi A_4 e^{-\beta y}\}. \quad (42)$$

Now we have to distinguish between two different conditions of symmetry, namely even symmetry and odd symmetry, as follows.

3.1.1. Even symmetry

In such a case, we have

$$u(y) = u(-y). \quad (71)$$

Hence, using (41) we get

$$A_1 e^{\alpha y} + A_2 e^{-\alpha y} + A_3 e^{\beta y} + A_4 e^{-\beta y} = A_1 e^{-\alpha y} + A_2 e^{\alpha y} + A_3 e^{-\beta y} + A_4 e^{\beta y},$$

or

$$(A_1 - A_2) \sinh \alpha y + (A_3 - A_4) \sinh \beta y = 0 \quad (72)$$

and

$$v(y) = v(-y). \quad (73)$$

Hence, using (42) we get

$$\begin{aligned} \psi A_1 e^{\alpha y} - \psi A_2 e^{-\alpha y} + \phi A_3 e^{\beta y} - \phi A_4 e^{-\beta y} \\ = \psi A_1 e^{-\alpha y} - \psi A_2 e^{\alpha y} + \phi A_3 e^{-\beta y} - \phi A_4 e^{\beta y}, \end{aligned}$$

or

$$(A_1 + A_2) \psi \sinh \alpha y + (A_3 + A_4) \phi \sinh \beta y = 0. \quad (74)$$

Hence, we must have from (72),

$$A_1 = A_2 \quad \text{and} \quad A_3 = A_4, \quad (75)$$

or from (74),

$$A_1 = -A_2 \quad \text{and} \quad A_3 = -A_4. \quad (76)$$

Notice that (75) satisfies (71) but not (73), while (76) satisfies (73) and not (71). Therefore, (71) and (73) can never be satisfied simultaneously, i.e., $u(y)$ and $v(y)$ can never be even simultaneously.

3.1.2. Odd symmetry

In such a case, we have

$$u(y) = -u(-y). \quad (77)$$

Hence, using (41) we get

$$A_1 e^{\alpha y} + A_2 e^{-\alpha y} + A_3 e^{\beta y} + A_4 e^{-\beta y} = -A_1 e^{-\alpha y} - A_2 e^{\alpha y} - A_3 e^{-\beta y} - A_4 e^{\beta y},$$

or

$$(A_1 + A_2) \cosh \alpha y + (A_3 + A_4) \cosh \beta y = 0 \quad (78)$$

and

$$v(y) = -v(-y). \quad (79)$$

Hence, using (42) we get

$$\begin{aligned} \psi A_1 e^{\alpha y} - \psi A_2 e^{-\alpha y} + \phi A_3 e^{\beta y} - \phi A_4 e^{-\beta y} \\ = -\psi A_1 e^{-\alpha y} + \psi A_2 e^{\alpha y} - \phi A_3 e^{-\beta y} + \phi A_4 e^{\beta y}, \end{aligned}$$

or

$$(A_1 - A_2) \psi \cosh \alpha y + (A_3 - A_4) \phi \cosh \beta y = 0. \quad (80)$$

Hence, we must have from (78),

$$A_1 = -A_2 \quad \text{and} \quad A_3 = -A_4, \quad (81)$$

or from (80),

$$A_1 = A_2 \quad \text{and} \quad A_3 = A_4. \quad (82)$$

But (81) satisfies (77) and not (79), while (82) satisfies (79) and not (77). Therefore, (77) and (79) can never be satisfied simultaneously, i.e., $u(y)$ and $v(y)$ can never be odd simultaneously.

As far as the pressure $p(y)$ is concerned, since the derivative of an even function is an odd function and vice versa,

$$p(y) = \frac{\delta u(y) - Dv(y)}{ibK}$$

is (i) even if $u(y)$ is even; (ii) odd if $u(y)$ is odd.

The above results can be summarized as follows.

(1) If $u(y)$, given by (65), is even, then $v(y)$, given by (66), is odd and $p(y)$, given by (67), is even.

(2) If $u(y)$, given by (68), is odd, then $v(y)$, given by (69), is even and $p(y)$, given by (70), is odd.

3.2. Verification of the boundary conditions

Let

$$\begin{aligned}\Delta_{\text{symmetric}} &= (\psi - \phi) \sinh(\alpha + \beta) + (\psi + \phi) \sinh(\alpha - \beta) = 0 \\ &= (\psi - \phi) \sinh \alpha \cosh \beta + (\psi - \phi) \cosh \alpha \sinh \beta \\ &\quad + (\psi + \phi) \sinh \alpha \cosh \beta - (\psi + \phi) \cosh \alpha \sinh \beta \\ &= 2(\psi \sinh \alpha \cosh \beta - \phi \cosh \alpha \sinh \beta)\end{aligned}$$

for the case where $u(y)$ and $p(y)$ are even and $v(y)$ is odd. Hence,

$$\begin{aligned}u(y) &= \frac{2A_1}{\cosh \beta} (\cosh \beta \cosh \alpha y - \cosh \alpha \cosh \beta y), \\ v(y) &= \frac{2\lambda A_1}{\cosh \beta} (\psi \cosh \beta \sinh \alpha y - \phi \cosh \alpha \sinh \beta y).\end{aligned}$$

Therefore,

$$\begin{aligned}u(1) &= \frac{2A_1}{\cosh \beta} (\cosh \beta \cosh \alpha - \cosh \alpha \cosh \beta) = 0, \\ u(-1) &= \frac{2A_1}{\cosh \beta} (\cosh \beta \cosh(-\alpha) - \cosh \alpha \cosh(-\beta)) \\ &= \frac{2A_1}{\cosh \beta} (\cosh \beta \cosh \alpha - \cosh \alpha \cosh \beta) = 0,\end{aligned}$$

and

$$\begin{aligned}v(1) &= \frac{2\lambda A_1}{\cosh \beta} (\psi \cosh \beta \sinh \alpha - \phi \cosh \alpha \sinh \beta) \\ &= \frac{2\lambda A_1}{\cosh \beta} \Delta_{\text{symmetric}} = 0, \\ v(-1) &= \frac{2\lambda A_1}{\cosh \beta} (\psi \cosh \beta \sinh(-\alpha) - \phi \cosh \alpha \sinh(-\beta)) \\ &= \frac{-2\lambda A_1}{\cosh \beta} (\psi \cosh \beta \sinh \alpha - \phi \cosh \alpha \sinh \beta) = \frac{-2\lambda A_1}{\cosh \beta} \Delta_{\text{symmetric}} = 0.\end{aligned}$$

Also, let

$$\begin{aligned}\Delta_{\text{antisymmetric}} &= (\psi - \phi) \sinh(\alpha + \beta) - (\psi + \phi) \sinh(\alpha - \beta) = 0 \\ &= (\psi - \phi) \sinh \alpha \cosh \beta + (\psi - \phi) \cosh \alpha \sinh \beta \\ &\quad - (\psi + \phi) \sinh \alpha \cosh \beta + (\psi + \phi) \cosh \alpha \sinh \beta \\ &= 2(\psi \cosh \alpha \sinh \beta - \phi \sinh \alpha \cosh \beta)\end{aligned}$$

for the case where $u(y)$ and $p(y)$ are odd and $v(y)$ is even. Hence,

$$u(y) = \frac{2A_1}{\sinh \beta} (\sinh \beta \sinh \alpha y - \sinh \alpha \sinh \beta y),$$

$$v(y) = \frac{2\lambda A_1}{\sinh \beta} (\psi \sinh \beta \cosh \alpha y - \phi \sinh \alpha \cosh \beta y).$$

Therefore,

$$u(1) = \frac{2A_1}{\sinh \beta} (\sinh \beta \sinh \alpha - \sinh \alpha \sinh \beta) = 0,$$

$$u(-1) = \frac{-2A_1}{\sinh \beta} (\sinh \beta \sinh \alpha - \sinh \alpha \sinh \beta) = 0,$$

and

$$v(1) = \frac{2\lambda A_1}{\sinh \beta} (\psi \sinh \alpha \cosh \alpha - \phi \sinh \alpha \cosh \beta)$$

$$= \frac{2\lambda A_1}{\sinh \beta} \Delta_{\text{antisymmetric}} = 0,$$

$$v(-1) = \frac{2\lambda A_1}{\sinh \beta} \Delta_{\text{antisymmetric}} = 0.$$

The above shows that $u(y)$ and $v(y)$ satisfy the boundary conditions as expected.

4. Numerical results and discussions

The complex eigenvalues $\delta = \delta_1 + i\delta_2$, where δ_1 and δ_2 are the attenuation and propagation parameters respectively, are computed and reported separately in [10].

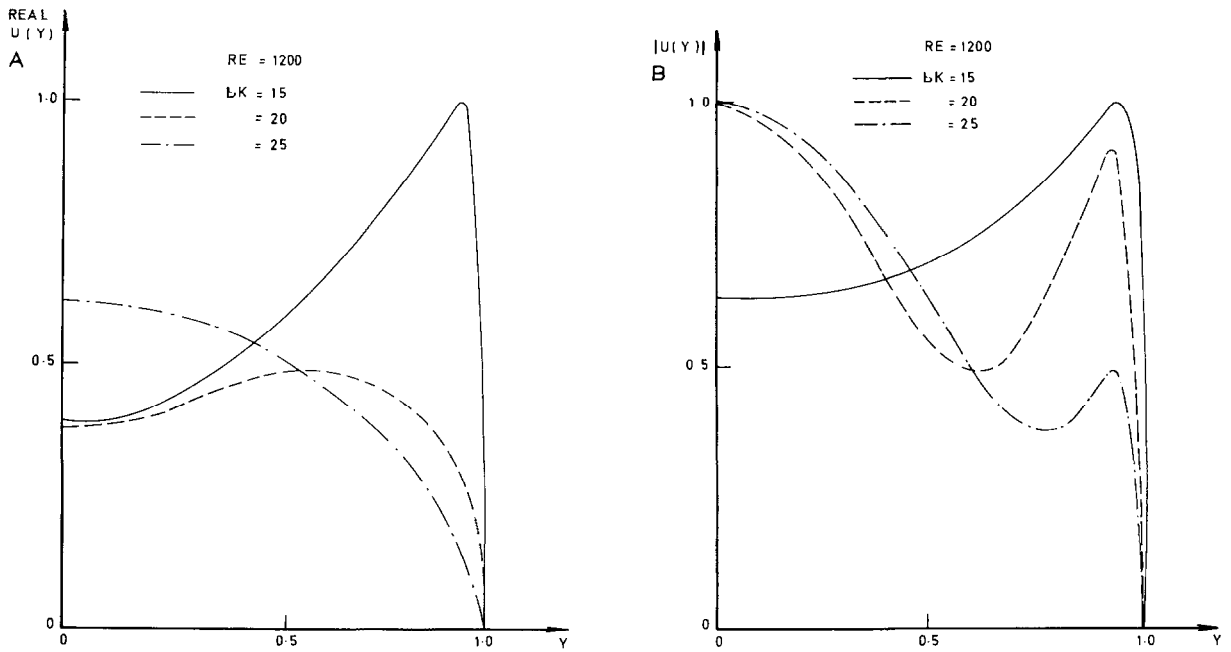
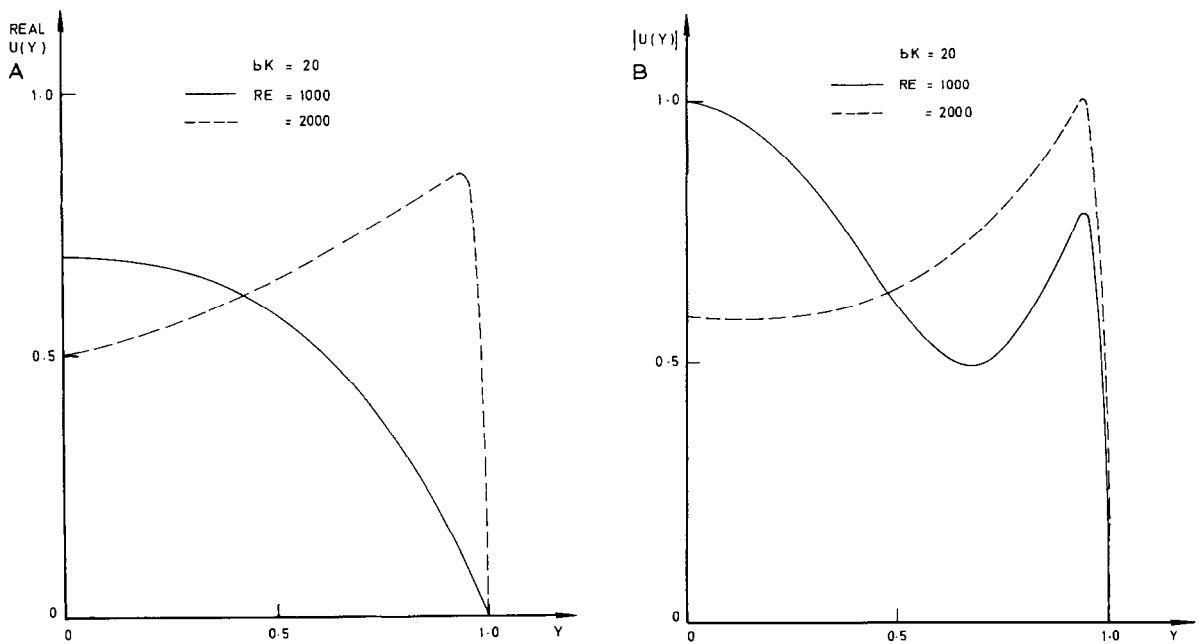
The complex eigenfunctions (or modal functions) u , v and p are computed for a wide range of the wave number bK and the Reynolds number Re and they are shown here for zeroth- and higher-order modes.

Although closed-form, exact solutions have been obtained as shown above, for u , v and p numerous numerical difficulties have been encountered. The sources of the difficulties are the existence of terms that involve $\exp(\alpha y)$ and $\exp(\beta y)$ which could result in overflow conditions for high values of Re . This happened in particular near the wall and near the duct centerline (i.e., near $y = 1$ and $y = 0$).

To overcome these numerical difficulties, the closed-form solution had to be rearranged differently for the cases of even and odd modes, and for u , v and p separately [4].

Examples of the obtained results are shown in Figs. 2–15.

For the zero mode, the effect of both bK and Re on the modal shape of the longitudinal velocity u is shown in Figs. 2–4. These results show the existence of a thin region near the duct wall through which the velocity u increases very rapidly from its zero value at the wall. This region is called the acoustic boundary layer in the literature.

Fig. 2(a). The effect of bK on the real component of u in the zero mode.Fig. 2(b). The effect of bK on the modal shape of u in the zero mode.Fig. 3(a). The effect of Re on the real component of u in the zero mode.Fig. 3(b). The effect of Re on the modal shape of u in the zero mode.

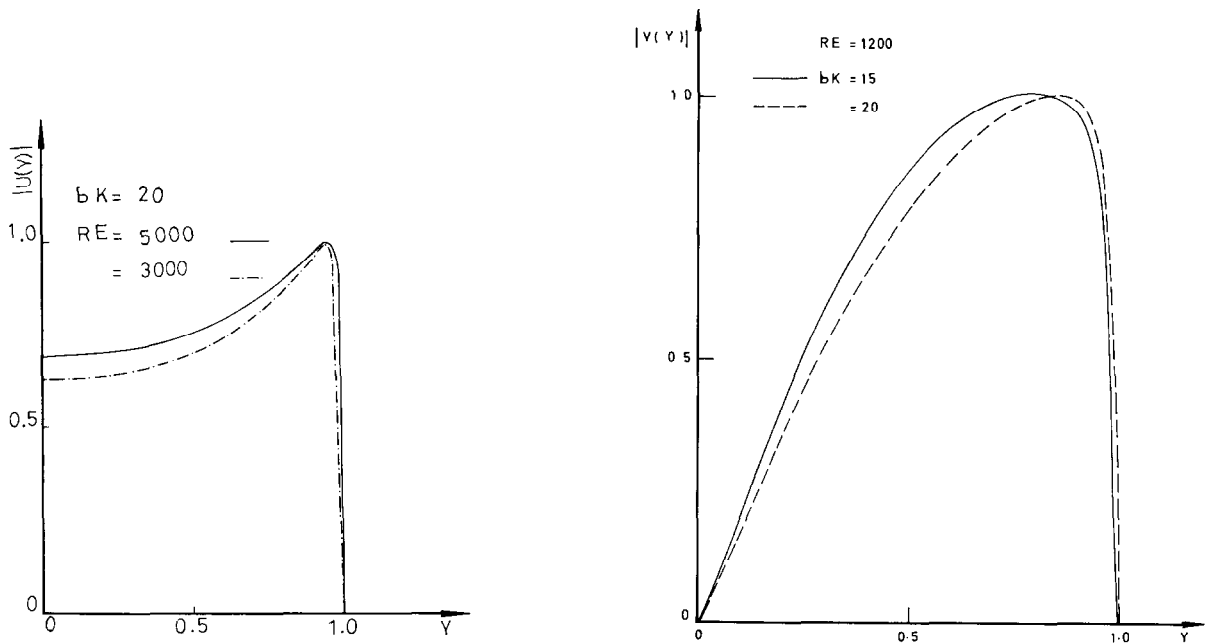


Fig. 4. The effect of Re (at high Re) on the modal shape of u in the zero mode.

Fig. 5. The effect of bK on the modal shape of v in the zero mode.

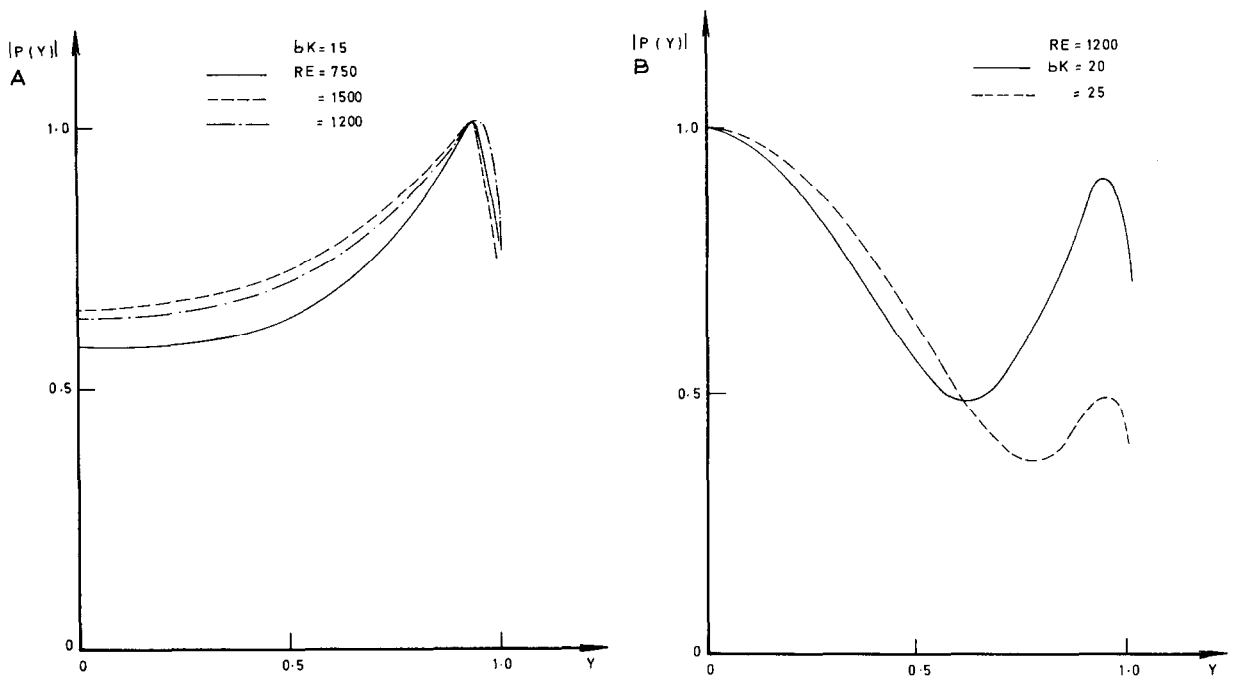
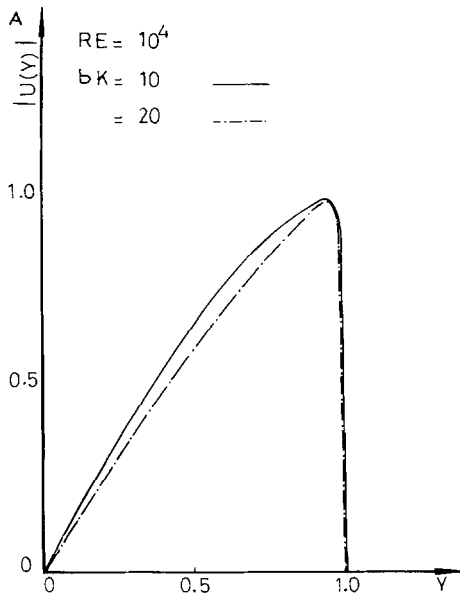
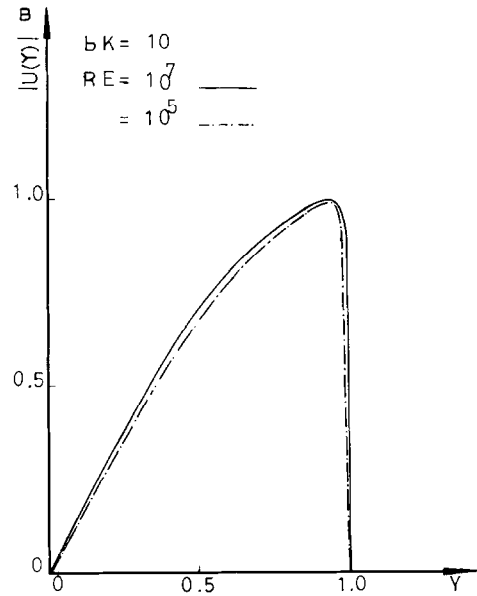
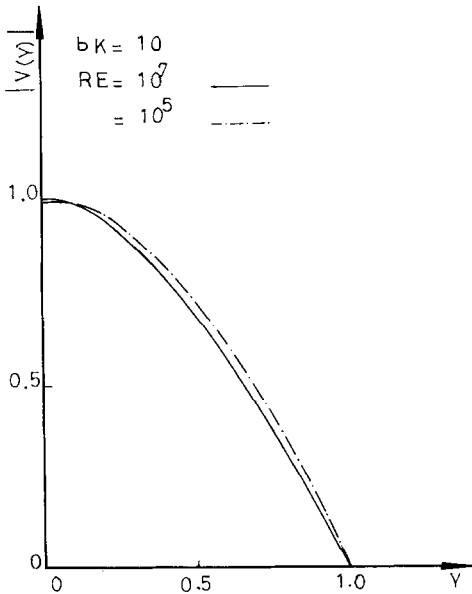
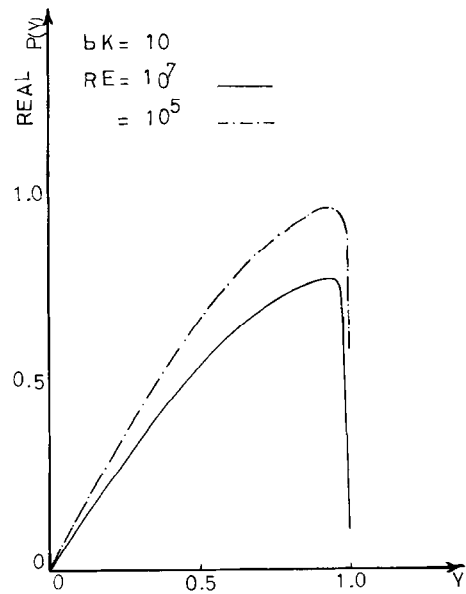


Fig. 6(a). The effect of Re on the modal shape of p in the zero mode.

Fig. 6(b). The effect of bK on the modal shape of p in the zero mode.

Fig. 7(a). The effect of bK on the modal shape of u in the first mode.Fig. 7(b). The effect of Re on the modal shape of u in the first mode.

It is clear that the modal shape (i.e., the distribution of u across the duct width) is far from the constant value (plane wave) resulting from the inviscid assumption. At the same Re , this deviation from the plane wave is more pronounced at higher bK as shown in Fig. 2(b). As

Fig. 8. The effect of Re on the modal shape of v in the first mode.Fig. 9. The effect of Re on the real component of p in the first mode.

shown in Figs. 3(b) and 4, at the same bK the modal shape gets closer to the plane wave as Re increases with the exception of the rapid drop to zero at the wall.

It is very interesting to note that the thickness of the acoustic boundary layer is almost constant and approximately equal to 0.06, independent of bK and Re . Also, it is worth noting that the gradient of u within the acoustic boundary layer is higher at higher Re and at lower bK .

Unlike the boundary layer of the mean flow in which the value of u is less than that in the core, the acoustic boundary layer can be defined as a region of high gradient of u .

The modal shape of the transverse velocity v of the zero mode is shown in Fig. 5. This is to be compared with zero value across the duct width for the inviscid case.

Figs. 6(a) and 6(b) show the effect of Re and bK on the modal shape of p in the zero mode respectively. The pressure drops within the boundary layer, but is nonzero at the wall. These modal shapes should be compared with the constant value in the inviscid case. As bK increases and Re decreases, the deviation from the constant value becomes more pronounced.

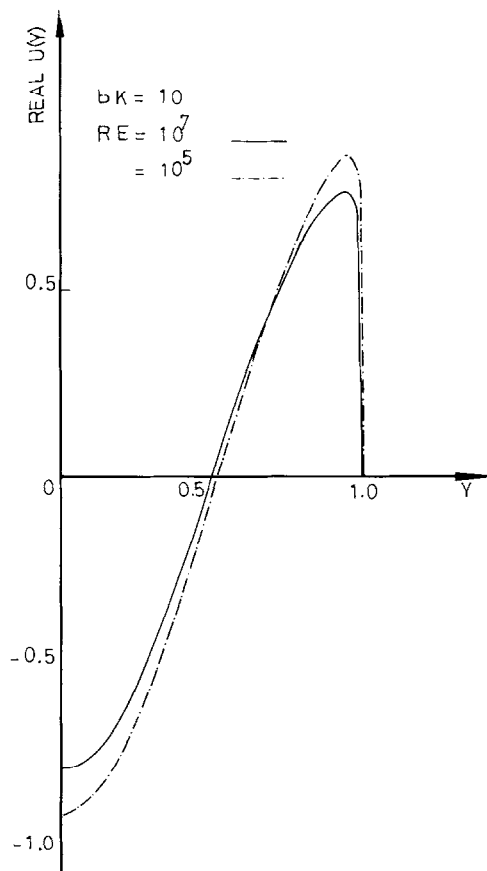


Fig. 10. The effect of Re on the real component of u in the second mode.

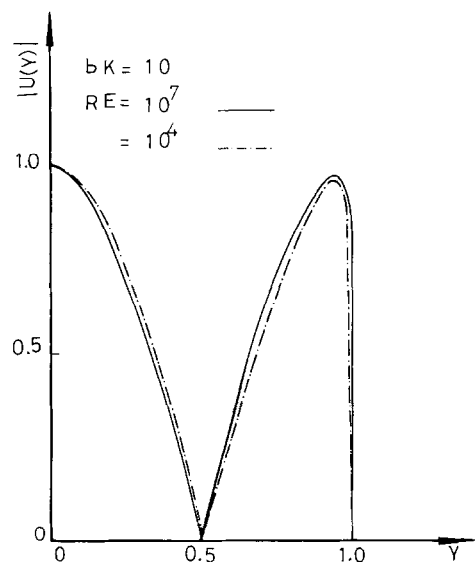
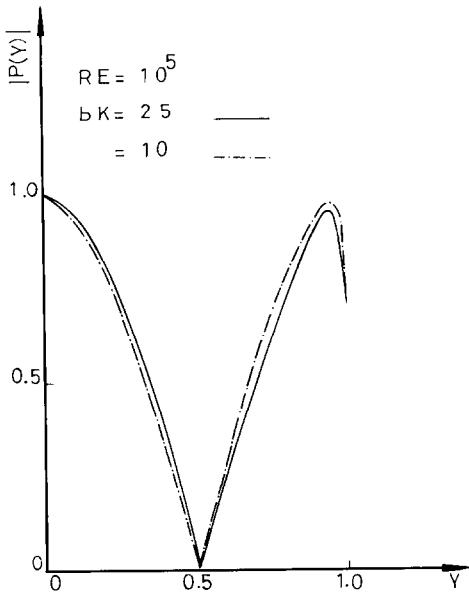
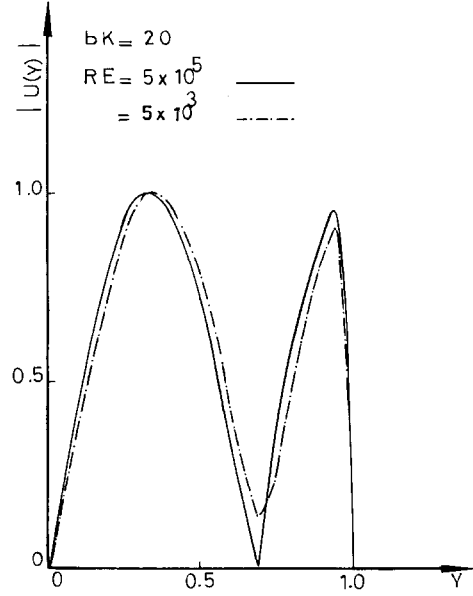
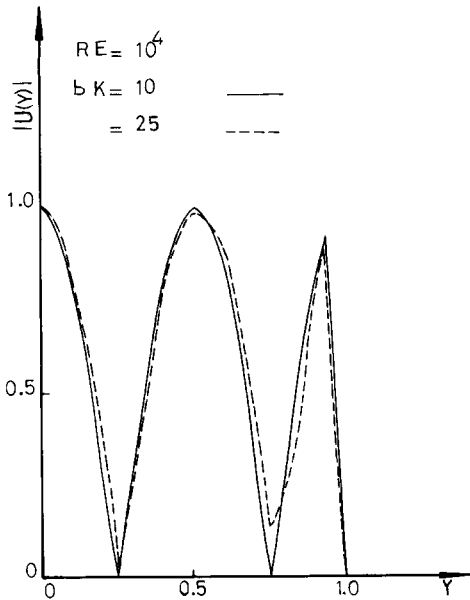
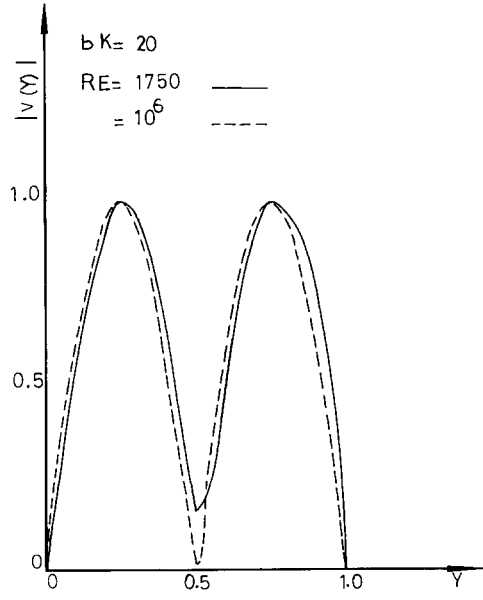


Fig. 11. The effect of Re on the magnitude of u in the second mode.

Fig. 12. The effect of bK on the magnitude of p in the second mode.Fig. 13. The effect of Re on the magnitude of u in the third mode.

The effect of bK and Re on the acoustic parameters u , v and p for the first and higher modes are shown in Figs. 7–15. The trends discussed in the case of zero mode prevail for the higher modes.

Fig. 14. The effect of bK on the magnitude of u in the fourth mode.Fig. 15. The effect of Re on the magnitude of u in the fourth mode.

5. Conclusions

- A mathematical model is constructed to describe the propagation of acoustic waves in a two-dimensional duct. The purpose has been to investigate the effect of the medium viscosity on the modal shapes of the wave parameters u , v and p , in particular, inside the thin region near the wall which is known as the acoustic boundary layer. The factors involved have been conveniently grouped into two nondimensional parameters bK and Re .
- For the case of no mean flow, closed-form, exact solutions have been obtained for u , v and p without imposing any assumptions concerning the thickness of the boundary layer or limiting bK or Re .
- The difficulties encountered in computing numerical values for u , v and p are solved and results are presented for the zeroth- and higher-order modes for a wide range of bK and Re .
- For the first time, the values of u , v and p are computed inside the acoustic boundary layer. The profile of these parameters are presented across the duct width.
- The results of this study do not support the widely used assumption that the effect of viscosity is limited to the thin boundary layer region and that the rest of the field is unaffected. The results presented here indicate that the distribution of the acoustic quantities across the duct width are strongly affected by the presence of viscosity.
- The acoustic boundary layer can be defined as the region near the wall where the longitudinal velocity changes very rapidly. It was found that the thickness of the acoustic boundary layer is almost constant and independent of the field parameters bK and Re .
- The effect of viscosity is stronger at low Re , as expected, and at high bK .
- The gradient of u , inside the boundary layer, is higher at higher Re and lower bK .

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